

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 4 Summer 2025 (Regular)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTCVC403 - STRUCTURAL MECHANICS - I

Time : 3 Hours]

[Total Marks : 60

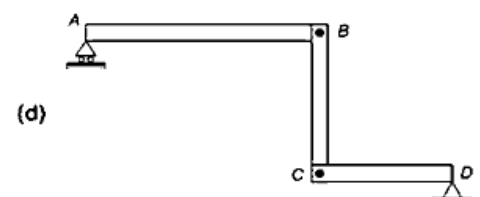
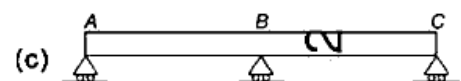
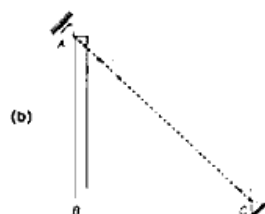
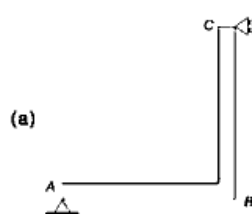
Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 The deflection at the free end of a cantilever beam subjected to a point load P at the free end is given by ----
a) $PL^3/3EI$ b) $PL^4/3EI$ c) $PL^3/6EI$ d) $PL^2/3EI$
- 2 A simply supported beam of length L carries a uniform load w per unit length. The deflection at the mid-span of the beam is ----
a) $5WL^4/384EI$ b) $WL^4/384EI$ c) $WL^3/48EI$ d) $WL^3/24EI$
- 3 In cantilever beams, slope and deflection at the fixed end are -----
Zero and Max. b) Zero, Zero c) Max. and Zero d) Max. and 1
- 4 If the member is denoted by m and joints by j , the condition for a frame to be redundant is given by
a) $m < 2j + 3$ b) $m < 2j - 3$ c) $m = 2j - 3$ d) $m > 2j - 3$
- 5 A plane frame has 15 members and 10 joints. It is fixed at both ends. The degree of static indeterminacy of the frame is -----
a) 2 b) 3 c) 6 d) 9
- 6 In the moment distribution method, the sum of distribution factors at any joint is always -----
a) = 0 b) = 1 c) > 1 d) < 1
- 7 In the moment distribution method, when considering a continuous beam, carry-over factor for the moment at the end of the beam is ---
a) 1.00 b) 0.5 c) 0.25 d) 0.75
- 8 The degree of external static indeterminacy for propped cantilever beam subjected to general loading is -----
a) 2 b) 3 c) 6 d) 9
- 9 Which one of the following structures is stable?



- 10 The slope-deflection method is best suited for analyzing -----
 a) Trusses b) Beams and frames with negligible joint translations
 c) Space frames d) Structures with significant joint translations
- 11 In the moment distribution method, when the distribution factor for a member is calculated, stiffness of the member is represented by---
 a) The flexural rigidity of the member b) The length of the beam
 c) The modulus of elasticity d) The deflection of the member
- 12 In moment distribution method, iterative process continues until ---
 a) The sum of moments at all joints is zero
 b) The unbalanced moments at all joints are within acceptable limits.
 c) The distribution factors at all joints become equal.
 d) The carryover factors at all joints become zero.

Q2. Solve the following.

- A) Enlist the different methods to find deflection in beams and explain in detail any one method. 6
 B) A cantilever beam of span 5 m is subjected to a concentrated load of 100 kN applied at the free end. Assume 6
 EI, the flexural rigidity of the beam, is constant. Calculate the maximum deflection at the free end of the beam
 in terms of EI.

Q3. Solve the following.

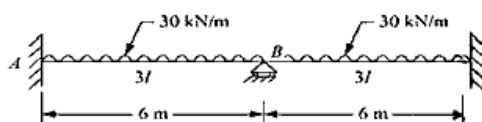
- A) Explain the concept of 'static & kinematic indeterminacy' with examples. 6
 B) Determine the deflection at the free end of a cantilever beam of length 5 m subjected to a point load of 100 kN at 6
 the free end. Use the strain energy method. Assume uniform flexural rigidity as 'EI.'

Q4. Solve Any Two of the following.

- A) Determine the reactions at the propped end for a propped cantilever beam of span 10 m subjected to a uniformly 6
 distributed load of 5 kN/m. Also find the bending moment at the fixed end.
 B) A fixed beam of span 'L' m is subjected to point load 'W' kN at the midspan. Draw shear force diagram (SFD) 6
 and bending moment diagram (BMD). Locate the point of contraflexure. Assume flexural rigidity EI is constant.
 Use a consistent deformation method.
 C) A fixed beam of span '10 m' is subjected to UDL '50' kN/m throughout the span. Find SFD and BMD. Locate 6
 the point of contraflexure. Flexural rigidity EI is constant. Use a consistent deformation method.

Q5. Solve Any Two of the following.

- A) State and explain the theorem of three moments. 6
 B) Explain the following terms: 6
 A) Carry over moment B) Carry factor C) Stiffness factor
 C) Analyze the two-span continuous beam ABC as shown in the figure below by the moment distribution method 6
 and draw the BMD & deflection profile.



Q6. Solve Any Two of the following.

- A) State and explain the assumptions made in the slope deflection method (SDM) while developing this method. 6
 B) Develop slope deflection equations for a two-span continuous beam ABC having each span (AB & BC) of 6
 length '4 m' subjected to a UDL of 10 kN/m. Support A is hinged, and B and C are rollers. Assume constant EI.
 C) Analyze the two-span continuous beam ABC by the slope deflection method, having both ends as fixed ones. 6
 Span AB is having 4 m length, M.I. as 'I,' and a subjected central point load of 40 kN. Span BC is having a 6 m
 length, M.I. as '2I,' and is subjected to a UDL of 20 kN/m over the entire span BC. Draw the BMD & elastic
 curve/deflection profile.